Brainstorming the solar radiation story

Divergent ideas, evidence questions, and early chart sketches before implementation.

Central audience question

How can Hong Kong planners understand when solar radiation is strongest, what weather conditions explain it, and when extreme opportunities occur?

Working hypotheses

H1 GSR and rainfall move in opposite directions.

H2 Sunshine duration is the strongest positive association; humidity is negative.

H3 Extreme GSR days cluster in high-sunshine, low-rainfall seasons.

Ideas removed or simplified

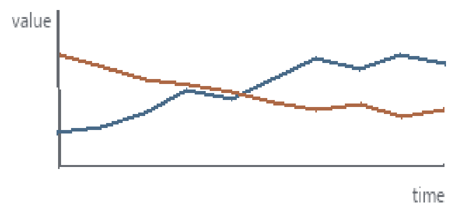
~~Map: one station cannot support a spatial claim~~

~~Raw daily line chart: too noisy across 11,887 days~~

~~Dense pairplot: too analytical for the target audience~~

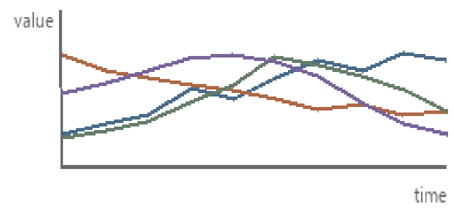
Candidate visual encodings

Annual GSR vs rainfall



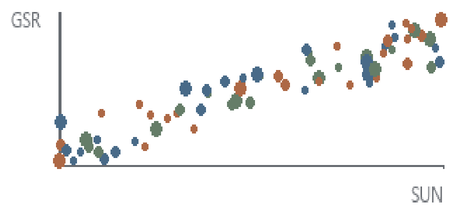
long-term context

Monthly climate rhythm



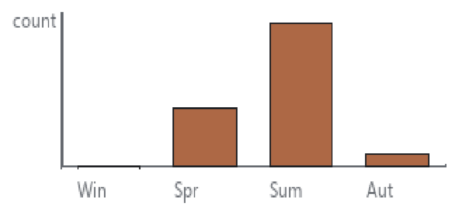
compare different units

SUN-GSR scatter



daily relationship

Extreme days by season



rare-event evidence

Possible interactions and narrative structure

Story steps

Move from context to explanation to action.

Year + season filters

Let planners focus without turning the story into a general dashboard.

Tooltips

Preserve raw values when standardised views are used.

Linked extreme date

Trace one summary record back to the daily scatterplot.

Alternative A: Scrollytelling climate story

A linear article-style experience where each scroll section introduces one visual claim.

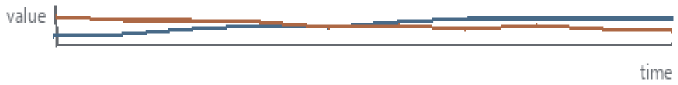
Complete interface sketch

WHEN SUNSHINE MEETS RAIN

A guided climate story for Hong Kong planners

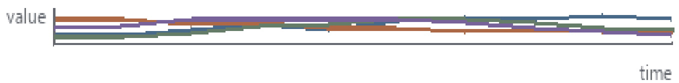
01 Long-term climate context

Short narrative paragraph explains the claim before the reader encounters the chart.



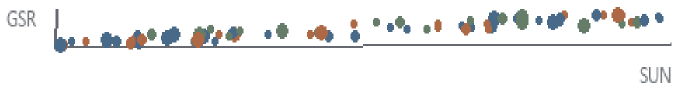
02 Seasonal rhythm

Short narrative paragraph explains the claim before the reader encounters the chart.



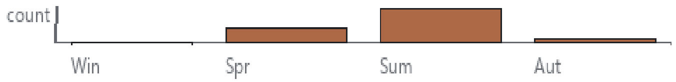
03 Daily sunshine relationship

Short narrative paragraph explains the claim before the reader encounters the chart.



04 Extreme solar days

Short narrative paragraph explains the claim before the reader encounters the chart.



Interaction sketch

Scroll activates each chart

The active section changes opacity and reveals its annotation. Hover provides exact values.

Strength

Very clear narrative sequence for a non-technical audience.

Risk

Readers cannot easily compare steps or return to a previous view while keeping filters.

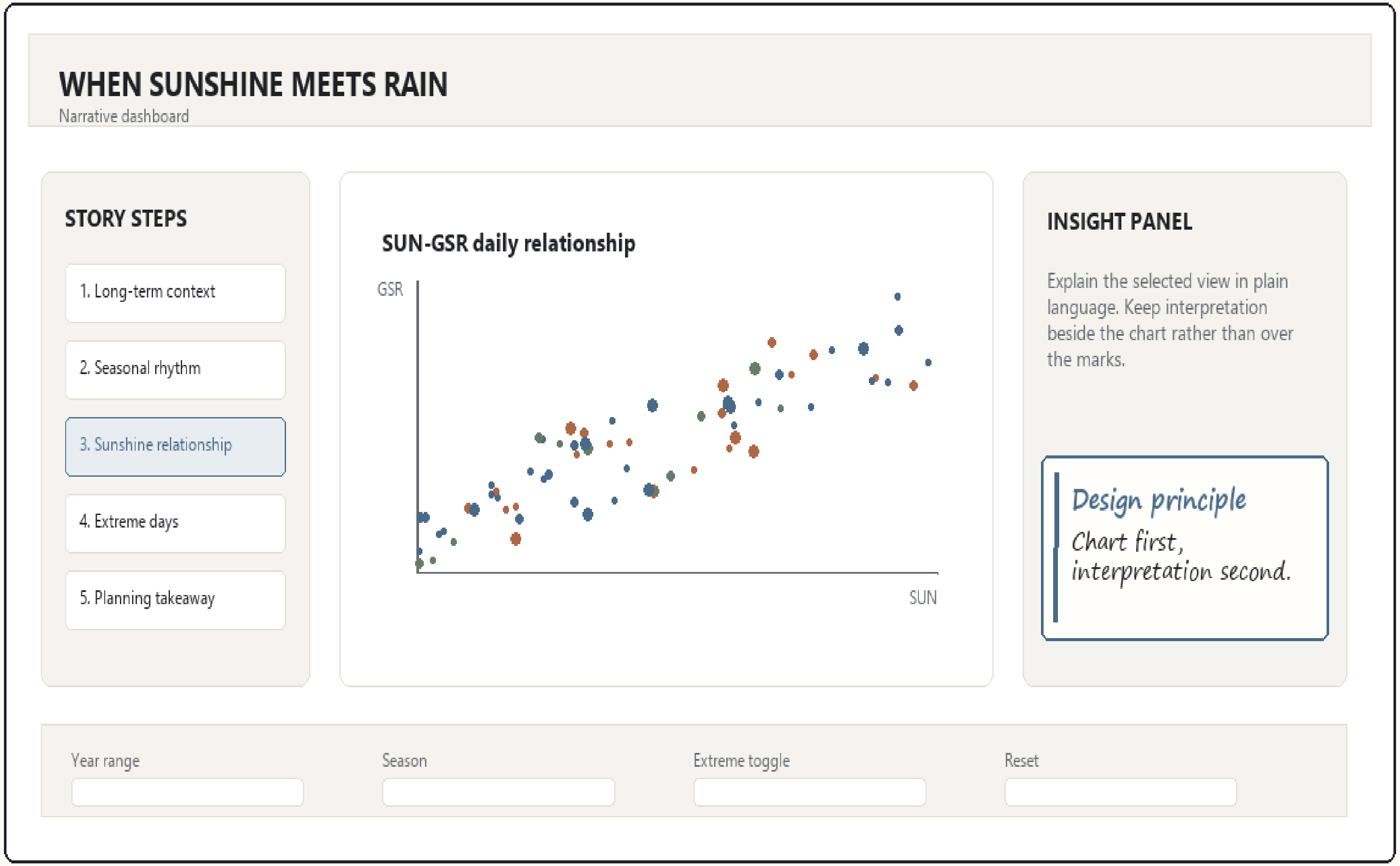
Decision

Keep the guided sequencing, but move away from a purely linear scroll so users can revisit evidence and use focused controls.

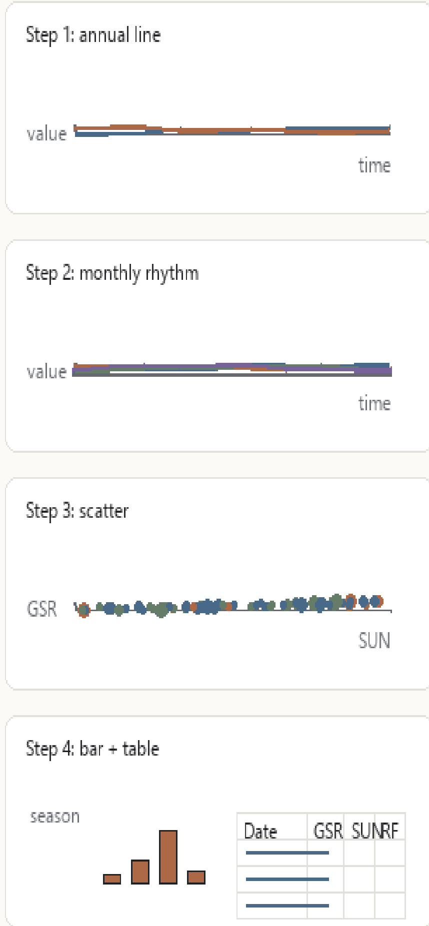
Alternative B: Guided story dashboard

A stable workspace with explicit story steps, a main chart, an insight panel, and focused controls.

Complete interface sketch



Step-specific chart sketches



Selected direction
Guided, but easy to revisit.

Alternative C: Seasonal and extreme-event explorer

A task-focused concept that prioritises monthly climate rhythm and rare high-radiation days.

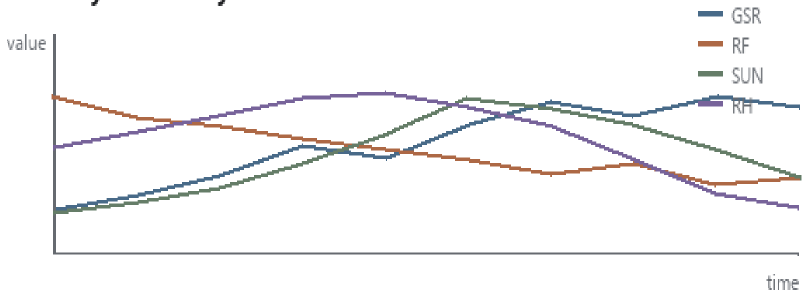
Complete interface sketch

SOLAR OPPORTUNITY EXPLORER

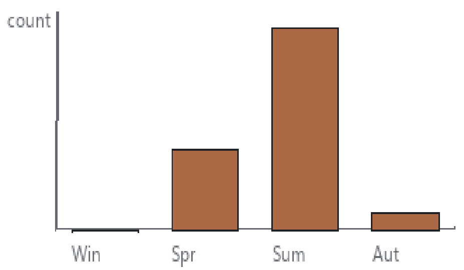
Seasonal patterns and extreme high-GSR days

Seasonal rhythm | Extreme days | Record details

Monthly climate rhythm



Extreme days by season



Extreme count	Average GSR	Average SUN	Average RF
595	25.79	10.85	0.47

Interaction and evidence notes

Season first

The concept makes monthly peaks, troughs, and summer clustering immediately visible.

Extreme evidence

Summary cards and a top-record table make rare events concrete rather than abstract.

Risk

It is weaker at explaining the full long-term story and the daily SUN-GSR relationship.

Decision

Carry the seasonal and extreme-event components into the guided dashboard from Sheet 3.

Final realisation: Guided five-step narrative

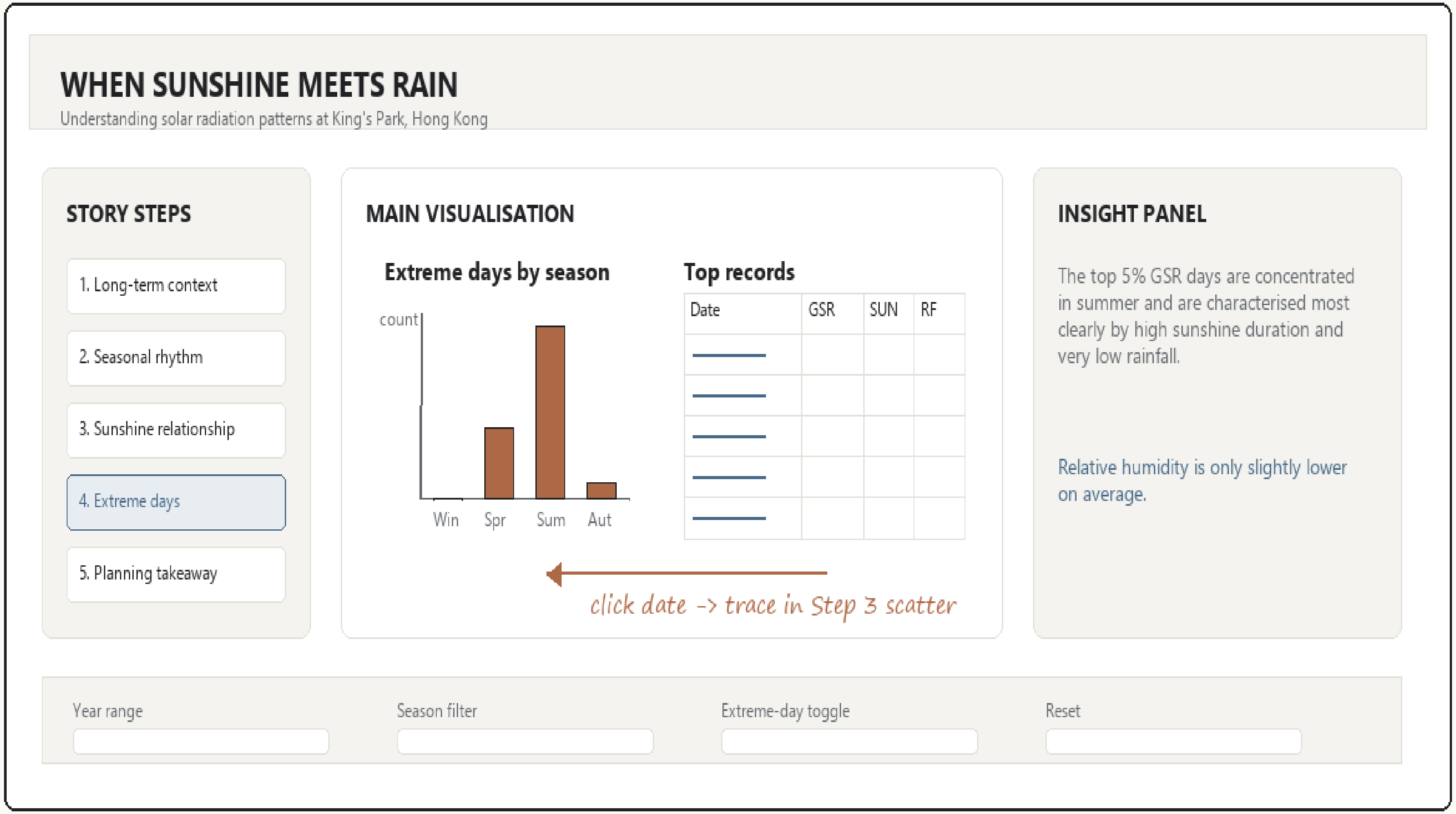
The implementable design combines Sheet 3's stable narrative workspace with Sheet 4's seasonal and extreme-event evidence.

Refinement after Part 1 feedback

Draft chart sketches, interaction arrows, and implementation annotations were added so the design communicates what each story step will actually show.

Evidence-aware refinement

The final wording does not overclaim humidity: extreme days are most clearly high-sunshine, low-rainfall summer events, while humidity is supporting context.



Five-step evidence plan

1. **Annual line** — Test H1 across years
2. **Monthly rhythm** — Show seasonal overlap
3. **SUN-GSR scatter** — Test H2 in daily data
4. **Bar + table** — Test H3 and trace records
5. **Takeaway cards** — Translate evidence for planners

Implementation

D3.js + local CSV + state.